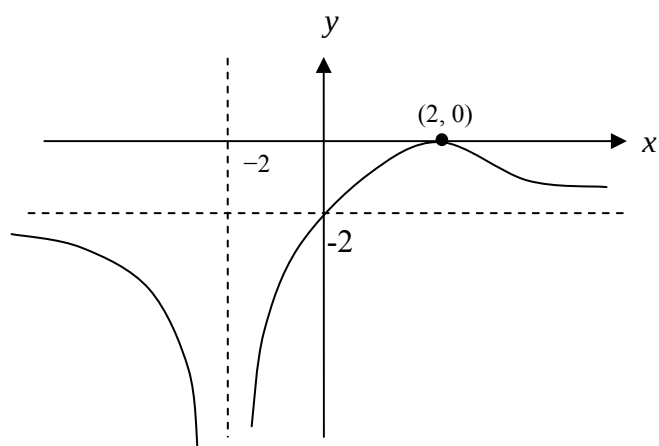
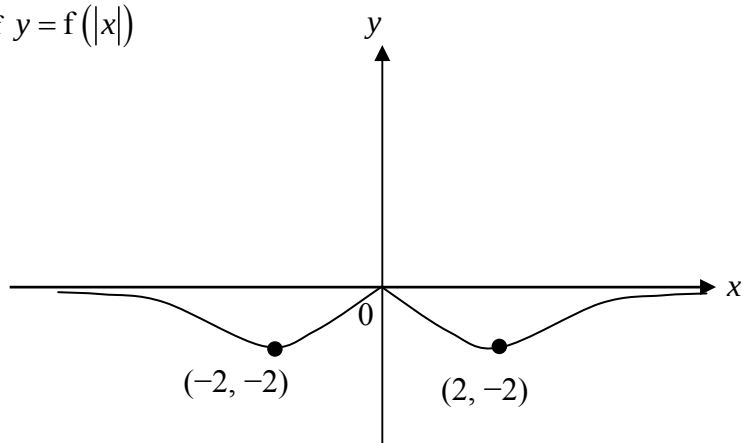


Solution to 2009 JJC Prelim Paper 1

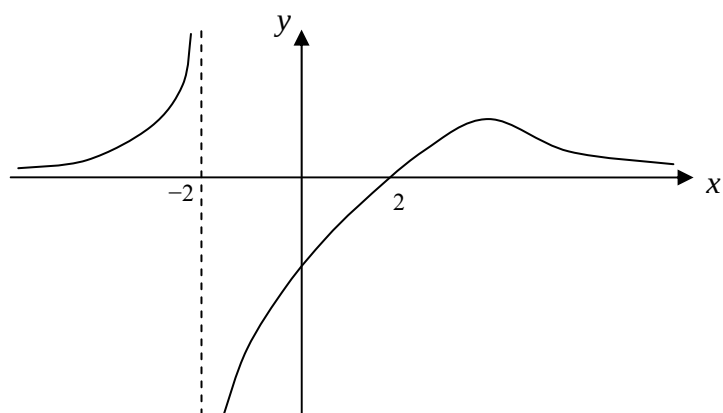
1(i) Graph of $y = -f(x) - 2$



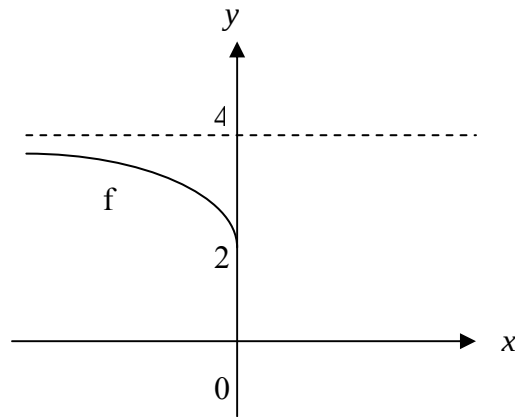
(ii) Graph of $y = f(|x|)$



(iii) Graph of $y = f'(x)$



2(i)



$$R_f = [2, 4)$$

(ii)

$$D_g = [0, 1) \cup (1, \infty)$$

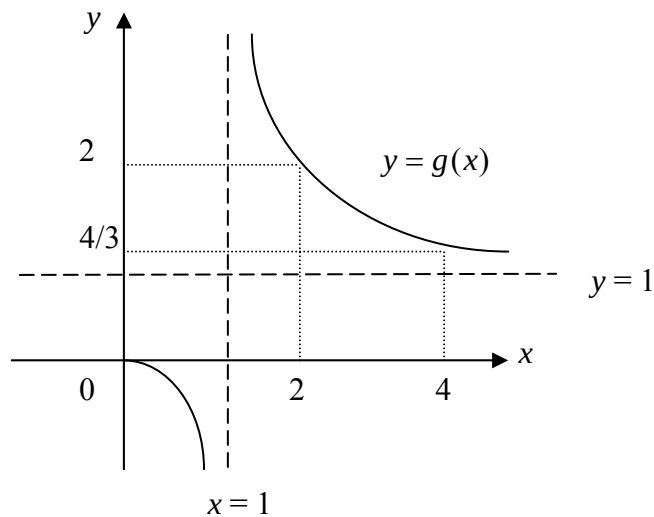
since $R_f = [2, 4)$ is a subset of $D_g = [0, 1) \cup (1, \infty)$

$\therefore$ gf exists. (shown)

$$gf(x) = g[f(x)] = g[4 - 2e^x] = 1 + \frac{1}{(4 - 2e^x) - 1} = 1 + \frac{1}{3 - 2e^x}$$

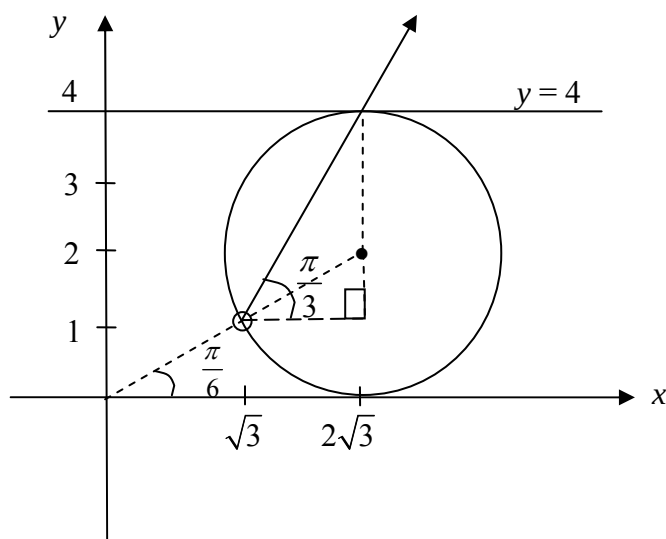
$$gf : x \mapsto 1 + \frac{1}{3 - 2e^x}, \quad x \in \mathbb{R}, \quad x \leq 0$$

(iii)



$$R_{gf} = \left(\frac{4}{3}, 2 \right]$$

3



$$z - z^* = 8i$$

$$x + iy - (x - iy) = 8i$$

$$2iy = 8i$$

$$y = 4$$

From the diagram, $w = 2\sqrt{3} + 4i$

4

$$\frac{dP}{dt} = 1\%(P) + 52(5)$$

$$\frac{dP}{dt} = 0.01P + 260 \quad (\text{Shown})$$

(i)

$$\int \frac{1}{0.01P + 260} dP = \int dt$$

$$\frac{1}{0.01} \ln(0.01P + 260) = t + c$$

$$P = 100(Ae^{0.01t} - 260)$$

When $t = 0$, $P = 1000$, $A = 270$

$$\text{Hence } P = 100(270e^{0.01t} - 260)$$

(ii)

$$\text{For } P > 2000, 100(270e^{0.01t} - 260) > 2000$$

$$e^{0.01t} > \frac{28}{27}$$

$$t > 3.64 \text{ years}$$

5(a)

$$\begin{aligned}\int \frac{1-4x}{4x^2+1} dx &= \int \left[\frac{1}{4x^2+1} - \frac{4x}{4x^2+1} \right] dx \\ &= \int \frac{1}{4x^2+1} dx - \int \frac{4x}{4x^2+1} dx \\ &= \frac{1}{2} \left[\tan^{-1} 2x \right] - \frac{1}{2} \ln(4x^2+1) + c\end{aligned}$$

(b)

$$x = \tan \theta \Rightarrow dx = \sec^2 \theta d\theta$$

$$\text{when } x = \infty, \tan \theta = \infty \Rightarrow \theta = \frac{\pi}{2}; \quad \text{when } x = 1, \tan \theta = 1 \Rightarrow \theta = \frac{\pi}{4}$$

$$\begin{aligned}\int_1^\infty \frac{x^2}{(x^2+1)^2} dx &= \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{\tan^2 \theta}{(\tan^2 \theta + 1)^2} \sec^2 \theta d\theta \\ &= \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{\tan^2 \theta}{\sec^2 \theta} d\theta \\ &= \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \sin^2 \theta d\theta \\ &= \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{1 - \cos 2\theta}{2} d\theta \\ &= \frac{1}{2} \left[\theta - \frac{1}{2} \sin 2\theta \right]_{\frac{\pi}{4}}^{\frac{\pi}{2}} \\ &= \frac{1}{2} \left[\frac{\pi}{2} - \left(\frac{\pi}{4} - \frac{1}{2} \right) \right] = \frac{1}{2} \left[\frac{\pi}{4} + \frac{1}{2} \right]\end{aligned}$$

6(a)

Let x be no. of chickens.Let y be no. of horses.Let z be no. of sheep.

$$z = 2x \quad \Rightarrow 2x + 0y - z = 0 \text{ -----(1)}$$

$$2x + 4y + 4z = 1250 \quad \Rightarrow 2x + 4y + 4z = 1250 \text{ -----(2)}$$

Case 1:

$$\text{If } x + y + z = 250 \text{ -----(3)}$$

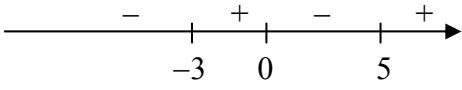
By GC, $x = -125$, $y = 625$, $z = -250$ (rejected)

(Alternative):

Reject $x + y + z = 250$, because any combination of 250 animals will never have 1250 legs.(Maximum no of legs = $250 \times 4 = 1000$)**Case 2:**

$$\text{If } x + y + z = 350 \text{ -----(3)}$$

By GC, $x = 75$, $y = 125$, $z = 150$

(b)	$\frac{2x-18}{x+3} - (x-6) > 0$ $\frac{2x-18-(x+3)(x-6)}{x+3} > 0$ $\frac{-x^2+5x}{x+3} > 0$ $\frac{x(x-5)}{x+3} < 0$ $x < -3 \quad \text{or} \quad 0 < x < 5$  $\frac{\ln x^2 - 18}{\ln x + 3} > \ln x - 6$ $\Rightarrow \frac{2 \ln x - 18}{\ln x + 3} > \ln x - 6$ For $\frac{2 \ln x - 18}{\ln x + 3} > \ln x - 6$, replace x with $\ln x$ $\ln x < -3 \quad \text{or} \quad 0 < \ln x < 5$ $0 < x < e^{-3} \quad \text{or} \quad 1 < x < e^5$
7(a)	Let P_n be the statement that "$u_n = \frac{1}{2}[7-5(3)^n]$" when $n = 0$, L.H.S. = $u_0 = 1$ (given) $\text{R.H.S.} = \frac{1}{2}[7-5(3)^0] = 1$ $\therefore P_0$ is true. Assume that P_k is true, i.e. $u_k = \frac{1}{2}[7-5(3)^k]$ When $n = k+1$, R.H.S. = $\frac{1}{2}[7-5(3)^{k+1}]$ L.H.S. = u_{k+1} $= 3u_k - 7 \text{ (given)}$ $= \frac{3}{2}[7-5(3)^k] - 7$ $= \frac{1}{2}[7-5(3)^{k+1}]$ $\therefore P_{k+1}$ is true Since P_0 is true and P_k is true $\Rightarrow P_{k+1}$ is true, By the principle of induction, P_n is true for all all non-negative integral values of n.

(b)

$$\begin{aligned}
\text{L.H.S.} &= \frac{1}{n!} - \frac{1}{(n+1)!} = \frac{1}{n!} - \frac{1}{n!(n+1)} \\
&= \frac{(n+1)-1}{n!(n+1)} \\
&= \frac{n}{n!(n+1)} \\
&= \frac{1}{(n-1)!n(n+1)} \\
&= \frac{1}{(n-1)!(n+1)} \\
&= \frac{1}{(n-1)!n + (n-1)!} \\
&= \frac{1}{n! + (n-1)!} = \text{R.H.S. (Shown)}
\end{aligned}$$

$$\begin{aligned}
&\sum_{n=1}^N \frac{1}{n! + (n-1)!} \\
&= \sum_{n=1}^N \left[\frac{1}{n!} - \frac{1}{(n+1)!} \right] \\
&= \left[\frac{1}{1!} - \frac{1}{2!} \right] + \left[\frac{1}{2!} - \frac{1}{3!} \right] + \left[\frac{1}{3!} - \frac{1}{4!} \right] + \dots + \left[\frac{1}{N!} - \frac{1}{(N+1)!} \right] \\
&= 1 - \frac{1}{(N+1)!}
\end{aligned}$$

Since $\frac{1}{n! + (n-1)!} > \frac{1}{n! + n!}$

$$\sum_{n=1}^N \frac{1}{n! + (n-1)!} > \sum_{n=1}^N \frac{1}{n! + n!}$$

$$\sum_{n=1}^N \frac{1}{n! + (n-1)!} > \sum_{n=1}^N \frac{1}{2n!}$$

$$\sum_{n=1}^N \frac{1}{n!} < 2 \sum_{n=1}^N \frac{1}{n! + (n-1)!}$$

$$\sum_{n=1}^N \frac{1}{n!} < 2 \left[1 - \frac{1}{(N+1)!} \right]$$

As $N \rightarrow \infty$, $\frac{1}{(N+1)!} \rightarrow 0$

$$\therefore \sum_{n=1}^{\infty} \frac{1}{n!} < 2 \text{ (shown)}$$

8	$A_4 - A_2 = G_3$ $(a + 3d) - (a + d) = ar^2$ $2d = ar^2 \Rightarrow d = \frac{ar^2}{2}$ $5A_5 - 6a = 9G_5$ $5(a + 4d) - 6a = 9ar^4$ $9ar^4 - 20d + a = 0 \quad \text{--- (1)}$ Sub $d = \frac{ar^2}{2}$ into (1) $9ar^4 - 20\left(\frac{ar^2}{2}\right) + a = 0$ $9ar^4 - 10ar^2 + a = 0$ $9r^4 - 10r^2 + 1 = 0 \quad \text{(Shown)}$ 8(i) $9r^4 - 10r^2 + 1 = 0$ $r^2 = 1 \quad \text{or} \quad r^2 = \frac{1}{9}$ $r = \pm 1 \text{ (rejected)} \quad \text{or} \quad r = -\frac{1}{3} \text{ (rejected)} \quad \text{or} \quad r = \frac{1}{3}$ Since $r = \frac{1}{3} < 1$, the geometric series is convergent. $S_\infty = \frac{a}{1-r} = \frac{a}{1-\frac{1}{3}} = \frac{3}{2}a$. (ii) $\sum_{n=1}^N A_n > 10 \sum_{n=1}^{\infty} G_n$ $\frac{N}{2} [2a + (N-1)d] > 10 S_\infty$ $d = \frac{ar^2}{2} = \frac{a}{18}$ $\frac{N}{2} \left[2a + (N-1) \left(\frac{a}{18} \right) \right] > 10 \left(\frac{3}{2}a \right)$ $N(N+35) > 540$ From the GC, the least value of $N=12$.
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9(i)

$$\overrightarrow{OA} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \quad \overrightarrow{OB} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} \quad \overrightarrow{OC} = \begin{pmatrix} 4 \\ 2 \\ 1 \end{pmatrix}$$

$$\overrightarrow{AB} = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} \quad \overrightarrow{AC} = \begin{pmatrix} 3 \\ 2 \\ 0 \end{pmatrix}$$

$$\underline{n} = \overrightarrow{AB} \times \overrightarrow{AC} = \begin{pmatrix} 2 \\ -3 \\ -1 \end{pmatrix}$$

$$r \cdot \underline{n} = a \cdot \underline{n}$$

$$r \cdot \begin{pmatrix} 2 \\ -3 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -3 \\ -1 \end{pmatrix}$$

$$2x - 3y - z = 1 \quad (\text{Shown})$$

9(ii)

Eqn of perpendicular line passing thru D to plane is

$$r = \begin{pmatrix} 5 \\ 1 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -3 \\ -1 \end{pmatrix} \quad \text{--- (1)}$$

$$\text{Eqn of plane} \quad r \cdot \begin{pmatrix} 2 \\ -3 \\ -1 \end{pmatrix} = 1 \quad \text{--- (2)}$$

$$\text{Sub (1) into (2)} \quad \begin{pmatrix} 5+2\lambda \\ 1-3\lambda \\ -1-\lambda \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -3 \\ -1 \end{pmatrix} = 1$$

$$\lambda = -\frac{1}{2}$$

$$\text{Foot of perpendicular :} \quad r = \begin{pmatrix} 5 \\ 1 \\ -1 \end{pmatrix} - \frac{1}{2} \begin{pmatrix} 2 \\ -3 \\ -1 \end{pmatrix} = \begin{pmatrix} 4 \\ 5/2 \\ -1/2 \end{pmatrix}$$

$$\text{Shortest Dist from D to plane :} \quad \left| \begin{pmatrix} 4 \\ 5/2 \\ -1/2 \end{pmatrix} - \begin{pmatrix} 5 \\ 1 \\ -1 \end{pmatrix} \right| = \left| \begin{pmatrix} -1 \\ 3/2 \\ 1/2 \end{pmatrix} \right| = \frac{\sqrt{14}}{2}$$

(iii)

If C and D are reflections, CD is a normal vector of Π_2

$$\overrightarrow{CD} = \begin{pmatrix} 5 \\ 1 \\ -1 \end{pmatrix} - \begin{pmatrix} 4 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ -2 \end{pmatrix}$$

If C and D are reflections, mid point of CD will exist in Π_2

$$\text{Mid point of } CD = \frac{1}{2} \begin{pmatrix} 5+4 \\ 1+2 \\ -1+1 \end{pmatrix} = \begin{pmatrix} 9/2 \\ 3/2 \\ 0 \end{pmatrix}$$

$$\text{Equation of } \Pi_2: r \cdot \begin{pmatrix} 1 \\ -1 \\ -2 \end{pmatrix} = \begin{pmatrix} 9/2 \\ 3/2 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -1 \\ -2 \end{pmatrix}$$

$$x - y - 2z = 3 \quad (\text{Shown})$$

$$\text{Using GC, eqn of intersection line is } r = \begin{pmatrix} 8 \\ 5 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 5 \\ 3 \\ 1 \end{pmatrix}, \quad \text{where } \lambda \in \mathbb{R}$$

10(a)

$$i(w+2) = z \quad \text{--- (1)}$$

$$wz = -4i \quad \text{--- (2)}$$

Subst (1) into (2),

$$wi(w+2) = -4i$$

$$w(w+2) = -4$$

$$(w)^2 + 2w + 4 = 0$$

$$w = -1 - i\sqrt{3} \quad \text{as } \text{Im}(w) < 0$$

$$z = \sqrt{3} + i$$

(b)

$$\text{Let } z^4 = -16i = 16e^{i\left(-\frac{\pi}{2}\right)}$$

$$\begin{aligned} z &= 2e^{i\left(\frac{-\frac{\pi}{2} + 2k\pi}{4}\right)} = 2e^{i\left(\frac{-\pi + 4k\pi}{8}\right)}, \quad k = -1, 0, 1, 2 \\ &= 2e^{i\left(-\frac{5\pi}{8}\right)}, 2e^{i\left(-\frac{\pi}{8}\right)}, 2e^{i\left(\frac{3\pi}{8}\right)}, 2e^{i\left(\frac{7\pi}{8}\right)} \end{aligned}$$

(c)

$$z = 1 - \sqrt{3}i = 2\left(\cos\left(-\frac{\pi}{3}\right) + i\sin\left(-\frac{\pi}{3}\right)\right)$$

$$z^n = 2^n\left(\cos\left(-\frac{n\pi}{3}\right) + i\sin\left(-\frac{n\pi}{3}\right)\right)$$

$$\text{Since } z^n \text{ is real, } \sin\left(-\frac{n\pi}{3}\right) = 0$$

$$\Rightarrow \frac{n\pi}{3} = k\pi \Rightarrow n = 3k, \text{ where } k \in \mathbb{Z}^+$$

11(a)	Since z^n is real and negative, $\cos\left(-\frac{n\pi}{3}\right) < 0$. Given that $n = 3k$, $\cos\left(\frac{3k\pi}{3}\right) = \cos(k\pi) < 0$ $\Rightarrow k$ has to be odd $\Rightarrow k = 2c - 1$, $c \in \mathbb{Z}^+$ $\Rightarrow$ Set of values of $n = \{n : n = 3(2c - 1), c \in \mathbb{Z}^+\}$ $x = 3t$, $y = \frac{3}{t}$ $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = -\frac{1}{t^2}$ when $t = 2$ $\frac{dy}{dx} = -\frac{1}{4}$ $x = 6$, $y = \frac{3}{2}$ and the gradient of the normal at P is 4. The equation of the normal is $y - \frac{3}{2} = 4(x - 6)$ $\Rightarrow 2y = 8x - 45$(1) Subst $x = 3t$, $y = \frac{3}{t}$ into (1) we have $2\left(\frac{3}{t}\right) = 8(3t) - 45$ $6 = 24t^2 - 45t \Rightarrow 8t^2 - 15t - 2 = 0$ $(8t + 1)(t - 2) = 0 \Rightarrow t = -\frac{1}{8}$ as $t \neq 2$
(b)	$V = \pi r^2 h \Rightarrow \frac{dV}{dr} = 2\pi r h$ $\frac{dV}{dt} = \frac{dV}{dr} \left(\frac{dr}{dt}\right) \Rightarrow 15 = 2\pi h r \left(\frac{dr}{dt}\right)$ $\Rightarrow \frac{dr}{dt} = \frac{15}{2\pi h r}$(1) $A = 2\pi r^2 + 2\pi r h \Rightarrow \frac{dA}{dr} = 4\pi r + 2\pi h$ $\frac{dA}{dt} = \frac{dA}{dr} \left(\frac{dr}{dt}\right) \Rightarrow 30 = (4\pi r + 2\pi h) \left(\frac{dr}{dt}\right)$ $\Rightarrow \frac{dr}{dt} = \frac{15}{2\pi r + \pi h}$(2) (1) = (2) $\Rightarrow \frac{15}{2\pi h r} = \frac{15}{2\pi r + \pi h}$ $\Rightarrow (2r + h) = 2hr$ $\Rightarrow r = \frac{h}{2(h-1)}$ Since $r > 0 \Rightarrow \frac{h}{2(h-1)} > 0 \Rightarrow h > 1$ (deduced)